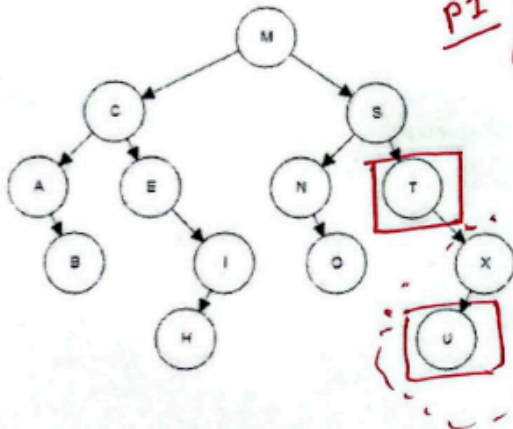


Q3. Do the following operations step by step in the below BST: [Mark-3]

- Identify the successor of node T- Provide a one-line explanation. [mark-1]
- Can node T be deleted using its successor? Give an explanation.[mark-1]
- After deleting node T, redraw the updated tree. [mark-1]

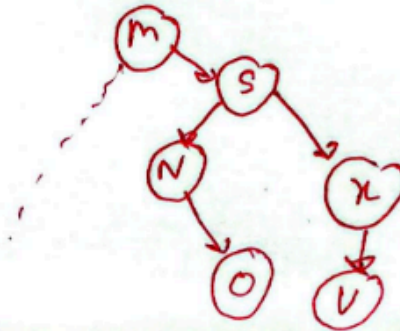


P1 Succ of T: right subtree, leftmost node, U
In order: L root R
 now A B C E H I M N O S
 (T) U X now.

P2: Node T has only 1 child. So no → Succ is only used when a node has 2 children.

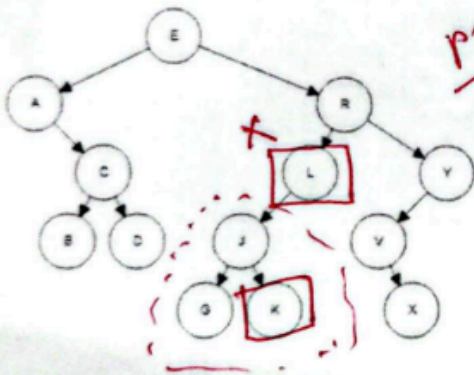
P3: After removal of T:

T has only 1 child
 So, parent & child connect.



Q3. Do the following operations step by step in the below BST: [Mark-3]

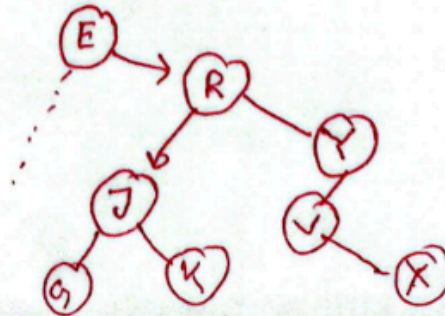
- P1 • Identify the predecessor of node L. Provide a one-line explanation. [mark-1]
P2 • Can node L be deleted using its predecessor? Give an explanation. [mark-1]
P3 • After deleting node L, redraw the updated tree. [mark-1]



P1 / Pred of L: left subtree right-most node, K
Inorder: L root R
none A B C D E G J K L R V
X Y none

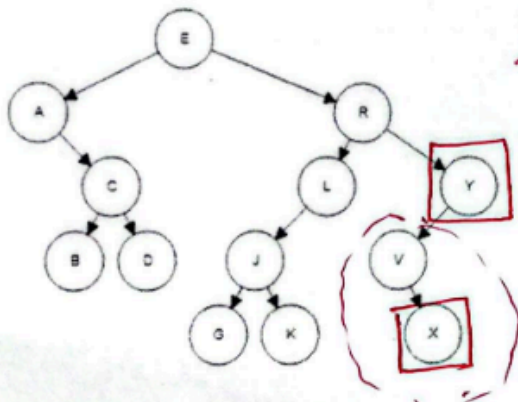
P2 / Node L has only 1 child. So, No → Pred is only used when a node has 2 children

P3 / After Removal of L:
L has 1 child.
→ So, parent & child connect.



Q3. Do the following operations step by step in the below BST: [Mark-3]

- P1 • Identify the predecessor of node Y. Provide a one-line explanation. [mark-1]
P2 • Can node Y be deleted using its predecessor? Give an explanation. [mark-1]
P3 • After deleting node Y, redraw the updated tree. [mark-1]

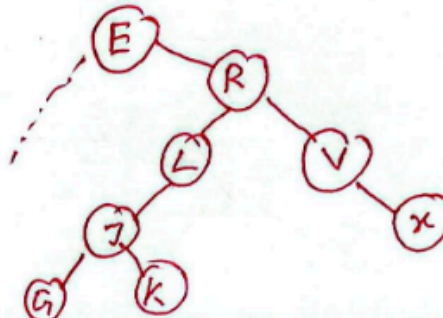


P1 Pred of Y: left sub-tree right most node, X
In order: L root R
 now A B C D E G J K L
 R V X Y none

P2: Node Y has only 1 child, so, No → Pred is only used when a node has 2 children.

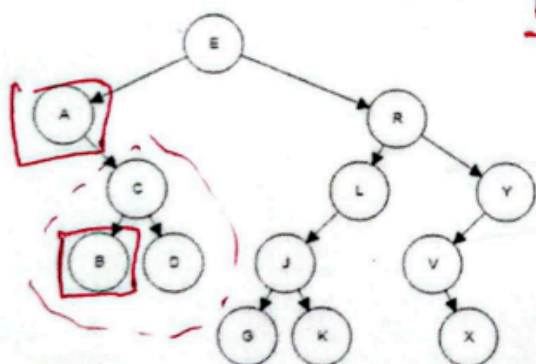
P3: After removal of Y:

Y has 1 child
 ↳ so, parent & child connect.



Q3. Do the following operations step by step in the below BST: [Mark-3]

- Identify the successor of node A- Provide a one-line explanation. [mark-1]
- Can node A be deleted using its successor? Give an explanation. [mark-1]
- After deleting node A, redraw the updated tree. [mark-1]



P1: Succ of A: right subtree, leftmost node, B
In order: L root R
 now A B C D E G J K L
 R V X Y now.

P2: Node A has only 1 child. So, No → Succ is only used when a node has 2 children.

P3: Removal of A:

A has 1 child
 → So, parent & child connect.

